Option TAQ

1 Data preview and preprocessing

1. Data preprocessing follows a standard guide by Andersen et al. [2021]

	ticker	conditions	correction	exchange	price	sip_timestamp	size	dt	date	underlying	expiry	cp_flag	strike
0	O:A251219C00130000	209.0	0	307	20.10	1764620209408000000	1	2025-12-01 20:16:49.408	2025-12-01	A	2025-12-19	C	130.0
1	O:A251219C00130000	209.0	0	309	20.10	1764620209408000000	1	2025-12-01 20:16:49.408	2025-12-01	A	2025-12-19	C	130.0
2	O:A251219C00135000	209.0	0	320	17.50	1764602484499000000	1	2025-12-01 15:21:24.499	2025-12-01	A	2025-12-19	C	135.0
3	O:A251219C00140000	227.0	0	300	11.36	1764611903918000000	1	2025-12-01 17:58:23.918	2025-12-01	A	2025-12-19	C	140.0
4	O:A251219C00140000	227.0	0	304	10.82	1764621699274000000	1	2025-12-01 20:41:39.274	2025-12-01	A	2025-12-19	C	140.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
7833006	O:ZYXI260220C00002500	227.0	0	312	0.45	1764600897366000000	6	2025-12-01 14:54:57.366	2025-12-01	ZYXI	2026-02-20	C	2.5
7833007	O:ZYXI260220C00002500	209.0	0	325	0.35	1764616838316000000	1	2025-12-01 19:20:38.316	2025-12-01	ZYXI	2026-02-20	C	2.5
7833008	O:ZYXI260220P00002500	209.0	0	325	1.55	1764621106983000000	12	2025-12-01 20:31:46.983	2025-12-01	ZYXI	2026-02-20	P	2.5
7833009	O:ZYXI260515C00002500	227.0	0	304	0.48	1764599945204000000	1	2025-12-01 14:39:05.204	2025-12-01	ZYXI	2026-05-15	C	2.5
7833010	O:ZYXI260515C00002500	209.0	0	308	0.45	1764605003830000000	1	2025-12-01 16:03:23.830	2025-12-01	ZYXI	2026-05-15	C	2.5

7833011 rows × 13 columns

Figure 1: trade_snapshots

2. Trade:
   (a) columns: ['ticker', 'conditions', 'correction', 'exchange', 'price', 'sip_timestamp', 'size', 'dt', 'date']
   (b) ticker → underlying, expiry, cp flag and strike price
      i. 4372 unique stocks
      ii. trade_date: 20251201 from sip_timestamp
      iii. maturity: from 20251205-20280112, AAPL as an example
      iv. strike price: divided by 1000
      v. eliminate adjusted contracts like O:ACB1260116C00000500[1]
   (c) sip_timestamp
      i. UTC timestamp → CT(Central time)
      ii. Elimination: trading session is within 8:30 CT and 15:00 CT for underlying stocks and until 15:15 CT for underlying ETF/ETP[2], align with 9:30 ET and 16:00 ET interval in Aït-Sahalia et al. [2025]

[1] `https://infomemo.theocc.com/infomemos?number=26853`

[2] Although some exchanges (e.g., BATS) accept early quotation and even run pre-market trading sessions, such records are typically more noisy and might cause time mismatching, if observations from several exchange markets are considered jointly.

(d) price and size: eliminate entries with zero transaction price and size

(e) **conditions**: trades with specific conditions which indicate a potentially irregular nature of a given entry, such as records of cancelled trades and non-firm quotes.

   i. Andersen et al. [2021] eliminate entries which correspond to canceled transactions (observations with condition codes 'A', 'C', 'E', 'G' and 'O')[3]

   ii. Our dataset contains [209., 227., 232., 233., 237., 240., 219., 231., 236., 235., 229., 228., 238., 239., 234., 204., 241., 242., 202., nan, 246., 248., 208.]

      A. not contains cancelled trades, eliminate nan

      B. however contains some late trades, may be detrimental to our setting[4]

(f) exchanges: our dataset contains 18 US exchanges [307, 309, 320, 300, 304, 312, 302, 301, 322, 316, 325, 308, 313, 323, 315, 319, 303, 318]

   i. non find the direct mapping[5]

   ii. For brevity, we only retain the underlyings that were constituents of the S&P 100 index on 20251201. Data source from CRSP

   iii. 78 stocks, 7.8m+ observations → 1.1m+ observations

3. Quotes: level-1 quote from different exchanges

	ticker	ask_exchange	ask_price	ask_size	bid_exchange	bid_price	bid_size	sequence_number	sip_timestamp
0	O:AAPL251205C00110000	312	170.60	1	312	165.60	1	759263	1764599400520234025
1	O:AAPL251205C00110000	303	168.65	1	303	166.00	1	759313	1764599400520455549
2	O:AAPL251205C00110000	303	168.65	1	303	165.75	1	759416	1764599400520770299
3	O:AAPL251205C00110000	309	170.15	1	312	166.10	1	875620	1764599400935108236
4	O:AAPL251205C00110000	307	169.50	1	307	166.95	1	883656	1764599400960994892
...	...	...	...	...	...	...	...	...	...
838752680	O:WMT261218P00160000	300	51.00	113	300	46.00	103	1594192102	1764622781708287073
838752681	O:WMT261218P00160000	300	51.00	111	313	46.00	102	1594216950	1764622781728324605
838752682	O:WMT261218P00160000	318	51.00	12	307	46.00	7	1594439279	1764622782483730613
838752683	O:WMT261218P00160000	316	51.00	101	316	46.00	101	1594542342	1764622782712536303
838752684	O:WMT261218P00160000	300	51.00	112	313	46.00	102	1596381173	1764622789549607796

838752685 rows × 9 columns

Figure 2: quotes_snapshots

(a) columns: ['ticker', 'ask_exchange', 'ask_price', 'ask_size', 'bid_exchange', 'bid_price', 'bid_size', 'sequence_number', 'sip_timestamp']

(b) For brevity, we only retain underlyings that belong to S&P 100, following Aït-Sahalia et al. [2025]

(c) ask_exchange and bid_exchange can be different in one observation

[3] `https://www.interactivebrokers.com.au/en/trading/include\_code\_OPRA.php`

[4] See details in `https://massive.com/glossary/trade-conditions`

[5] `https://massive.com/docs/llms.txt`

(d) sequence number

(e) 838.8+ m observations, aapl as an example, 57.4+ m observations.

(f) The experiment setup varies across stocks(time clocks and etc) and perform individual fitting rather than universal in Sirignano and Cont [2021]

4. **Merge scheme**: The best bid and ask information of trades are determined by the most recent quote information as of the time of the transaction. Eliminate

	ticker	conditions	correction	exchange	price	trade_ts	size	dt	date	underlying	...	trade_dt	ask_exchange	ask_price	ask_size	bid_exchange	bid_price	bid_size	sequence_nur
0	O:AAPL251205P00265000	209.0	0	323	0.23	1764599400003000000	21	2025-12-01 14:30:00.003	2025-12-01	AAPL	...	2025-12-01 14:30:00.003	323.0	0.27	6.0	323.0	0.21	1.0	7.339220
1	O:AAPL251212P00262500	209.0	0	323	0.58	1764599400022000000	4	2025-12-01 14:30:00.022	2025-12-01	AAPL	...	2025-12-01 14:30:00.022	323.0	0.61	2.0	323.0	0.51	8.0	7.813180
2	O:AAPL251205P00267500	209.0	0	323	0.35	1764599400023000000	12	2025-12-01 14:30:00.023	2025-12-01	AAPL	...	2025-12-01 14:30:00.023	323.0	0.39	1.0	323.0	0.29	1.0	7.341460
5	O:AAPL251212C00282500	209.0	0	323	2.57	1764599400125000000	25	2025-12-01 14:30:00.125	2025-12-01	AAPL	...	2025-12-01 14:30:00.125	309.0	5.10	1.0	309.0	0.10	1.0	7.832910
6	O:AAPL251205P00267500	227.0	0	323	0.37	1764599400255000000	1	2025-12-01 14:30:00.255	2025-12-01	AAPL	...	2025-12-01 14:30:00.255	323.0	0.39	1.0	323.0	0.29	1.0	7.341460
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
122071	O:AAPL260220P00260000	209.0	0	316	4.64	1764622799593000000	1	2025-12-01 20:59:59.593	2025-12-01	AAPL	...	2025-12-01 20:59:59.593	301.0	4.65	23.0	313.0	4.55	16.0	1.826452
122072	O:AAPL251219C00310000	227.0	0	300	0.16	1764622799616000000	1	2025-12-01 20:59:59.616	2025-12-01	AAPL	...	2025-12-01 20:59:59.616	301.0	0.16	10.0	320.0	0.15	2.0	1.834048
122073	O:AAPL251226C00280000	227.0	0	309	8.42	1764622799667000000	1	2025-12-01 20:59:59.667	2025-12-01	AAPL	...	2025-12-01 20:59:59.667	318.0	8.55	10.0	300.0	8.35	3.0	1.827062
122074	O:AAPL251205P00280000	209.0	0	303	1.26	1764622799885000000	1	2025-12-01 20:59:59.885	2025-12-01	AAPL	...	2025-12-01 20:59:59.885	320.0	1.27	9.0	318.0	1.23	1.0	1.834876
122075	O:AAPL251212C00285000	209.0	0	303	3.45	1764622799970000000	1	2025-12-01 20:59:59.970	2025-12-01	AAPL	...	2025-12-01 20:59:59.970	325.0	3.45	4.0	315.0	3.40	4.0	1.826515

122072 rows × 26 columns

Figure 3: TAQ_AAPL_snapshots

(a) Entries with zero price or size

(b) Entries with a negative spread

(c) Entries for which both bid and offer sizes do not exceed 1 contract

(d) Entries for which the spread exceeds 50 median spreads on that day

(e) Entries for which the mid-quote is by more than 10 mean absolute deviations apart from the rolling centered median (that is based on a rolling window of 50 observations)

(f) Entries for which the ratio of offer price to bid price exceeds 5 when the bid price is non-zero

(g) Entries for which a transaction price is either lower than the current bid price minus the current spread, or higher than the current offer price plus the current spread

(h) **Records with identical timestamps**: 32409 observation in AAPL[6]

   i. For trade, replaced with a single entry with the median transaction price and the total size.

   ii. For quote, replaced with a single entry with the median bid and ask prices and the total sizes

[6]quotes with identical millisecond stamps (F6) is substantial and varies from 3% to 12% of all quote entries across the underlyings in Andersen et al. [2021]

(i) from 122072 observations to 99405 observations, 1247 unique ticker, around 80 trades per ticker per day?

(j) we should only retain active contracts like near maturity and near ATM contracts

   i. near maturity < 14 days

   ii. (strike / spot(281.10) - 1).abs() < 5%

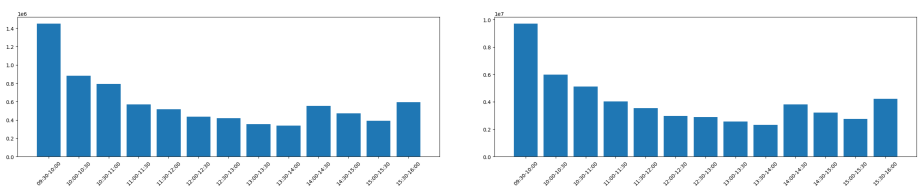
   iii. 48 unique tickers, 58583 observation

Findings

1. the quote-to-trade ratio is way higher than stock market. Before filters, apple contracts has 1284 unique contracts, only 122,076 transactions but 57,391,985 quotes[7]

2. trade activity is intense for ultra short-term contracts.

3. ATM trading dominates across all underlyings and maturity terms, while quotes varies: quote activity for OTM sometimes dominates



(a) Number of transactions (b) Total transaction size

Figure 4: Intraday distribution of option transactions and traded size(SP100)

[7]This is particularly true for actively traded equity indices and stocks. For example, during the period from January to August 2015, for Apple, we observe on average 271,641 trades and 3,283,067 quotes per day, implying an average quote-to-trade ratio of 12.1. This activity in the spot market is accompanied by an activity in the option market, amounting, on average, to 77,849 trades and 226,822,053 quotes per day across all 1,169 option classes written on Apple, suggesting an average quote-to-trade ratio of 2,913.6 for Apple options.

2 Response variables and Predictors

2.1 Response variables Y

First, notations

1. Time clocks, define a half open half closed interval. Just like bar in stocks, such that we can define OHLCV

$$\mathrm{Int}(T_1, T_2) = \{t \in R : T_1 < t \leq T_2\}. \tag{1}$$

   (a) canlendar: Δ measures the target time horizon over which the prediction takes place.

   (b) transaction: Δ measures the target number of transactions

   (c) volume: Δ measures the target volume

$$\mathrm{Int}^{\mathrm{forward}}(T, \Delta, \mathrm{M}) = \begin{cases} \mathrm{Int}(T, T+\Delta) & \text{if } M = \text{calendar} \\ \left\{t > T : \left(\sum_{s \in \mathrm{Int}(T,t)} \mathbb{1}_{\{V_s > 0\}}\right) \leq \Delta\right\} & \text{if } M = \text{transaction} \\ \left\{t > T : \left(\sum_{s \in \mathrm{Int}(T,t)} V_s\right) \leq \Delta\right\} & \text{if } M = \text{volume} \end{cases}.$$

Figure 5: time_clocks

2. Only under intervals, can we define quantities like return and etc.

3. **Since the timestamp of trade and quote are not one2one mapping**, denote $\mathbf{D^{txn}}$ as trade, and $\mathbf{D^{qt}}$ as quote counterpart.

$$\mathbf{D} = \mathbf{D^{txn}} \cup \mathbf{D^{qt}} \tag{2}$$

Then, formally define variables

1. Transaction return: average return from transactions in a forward window. At time $T \in \mathbf{D}$, with span Δ and clock mode M

$$\mathrm{Return}(T, \Delta, M) = \mathrm{Average}\left[P_t^{\mathrm{txn}} : t \in \mathbf{D}^{\mathrm{txn}} \cap \mathrm{Int}^{\mathrm{forward}}(T, \Delta, M)\right] / P_T - 1. \tag{3}$$

   (a) in Bali et al. [2023], the daily option return is defined as delta-hedged option gains, as the value of a self-financing portfolio consisting of a long option, hedged by a position in the underlying such that the portfolio is locally immune to changes in the stock price.

2. Price direction. At time $T \in \mathbf{D}$, with span Δ and clock mode M

$$\mathrm{Direction}(T, \Delta, M) = 1_{\left\{\mathrm{Return}(T,\Delta,M) > \bar{R}(\Delta,M)\right\}}. \tag{4}$$

(a) Unintuitively though, $\bar{R}(\Delta, M)$ close to 0, be the average past transaction return of the stock.

3. Transaction duration. At time $T \in \mathbf{D}$, with span Δ and clock$M \in$ {transaction, volume}

$$\text{Duration}(T, \Delta, M) = argmax_{t \in \mathbf{D}} \left\{ t \in \text{Int}^{\text{forward}}(T, \Delta, M) \right\} - T. \quad (5)$$

(a) measures the amount of (calendar) time it takes to record either Δ transactions or Δ shares

2.2 Predictors X

1. Disjoint windows like

(a) Canlendar clocks: 9 spans are

$$(\Delta_1, \Delta_2) \in \{(0, 0.1), (0.1, 0.2), (0.2, 0.4), \ldots, (12.8, 25.6)\},$$

(b) Transaction clocks: 9 spans are

$$(\Delta_1, \Delta_2) \in \{(0, 1), (1, 2), (2, 4), \ldots, (128, 256)\},$$

(c) Volume clocks:

$$(\Delta_1, \Delta_2) \in \{(0, 100), (100, 200), (200, 400), \ldots, (12800, 25600)\},$$

$$\text{Int}^{\text{back}}(T, \Delta_1, \Delta_2, \text{M}) = \begin{cases} \text{Int}(T - \Delta_2, T - \Delta_1) & \text{if } M = \text{calendar} \\ \left\{ t : t \leq T, \Delta_1 \leq \left(\sum_{s \in \text{Int}(t,T)} \mathbb{1}_{\{V_s > 0\}} \right) < \Delta_2 \right\} & \text{if } M = \text{transaction} \\ \left\{ t : t \leq T, \Delta_1 \leq \left(\sum_{s \in \text{Int}(t,T)} V_s \right) < \Delta_2 \right\} & \text{if } M = \text{volume} \end{cases} \cdot$$

2. 13 main predictors, each being implemented over the 9 time spans and 3 time clocks.

3. **need to alter it in option data**

(a) in AAPL contracts, about 48ticker and 58000+ successful transactions(1000+ ts per ticker per day), way lower than stocks

(b) thus it is necessary to design new time clocks in our setting

i. calendar clocks 5second spans(3min): SPANS = [(0, 5), (5, 10), (10, 20), (20, 40), (40, 80), (80, 120), (120, 180), (180, 240), (240, 300)]

ii. transaction clock xxx

iii. volume clock xxx

3 Training Scheme

1. Model: Lasso as an example
2. Estimation and forecast timing, following Chinco et al. [2019]
   (a) first estimate a model using data from the previous 30 minutes
   (b) apply the estimated coefficients to the most recent data
   (c) refit the model every 5 minutes
3. Fitting scheme
   (a) here AAPL as an example, since it is the most actively traded option classes
   (b) individual fitting here is necessary, because transaction clock and volume clock interval varies across options, not like 1min, 5min bars.
   (c) but pool fitting using canlendar clocks may be plausible [Sirignano and Cont, 2021]
4. why works?
   (a) "We show that approximately 80% of the overall predictability is achieved by relying on the most recent 10 milliseconds"

Table 1: Model-Specific Hyperparameters

Model	**Key Parameters**
Lasso	LassoCV: n_alphas=50, max_iter=20000, cv=TimeSeriesSplit(3)
Ridge	RidgeCV: alphas=logspace(−3,3,50), cv=TimeSeriesSplit(3)
ElasticNet	ElasticNetCV: cv=TimeSeriesSplit(3), random_state=42
RandomForest	n_estimators=100, max_depth=8, min_samples_leaf=20
XGBoost	n_estimators=100, max_depth=4, learning_rate=0.1

4 Results

1. OOS R^2 about 6.7% greater than daily option return around 2% in Bali et al. [2023]

Table 2: Model Performance Comparison

Model	n	RMSE	R^2	Dir. Acc.
RandomForest	27909	0.013031	0.143793	0.630119
ElasticNet	27909	0.013592	0.068439	0.616683
Lasso	27909	0.013598	0.067613	0.616253
XGBoost	27909	0.014399	-0.045425	0.618295
Ridge	27909	0.014695	-0.088793	0.615178

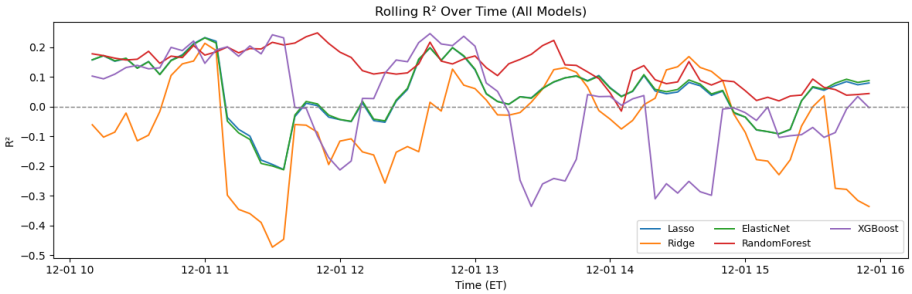


Figure 6: Rolling R2

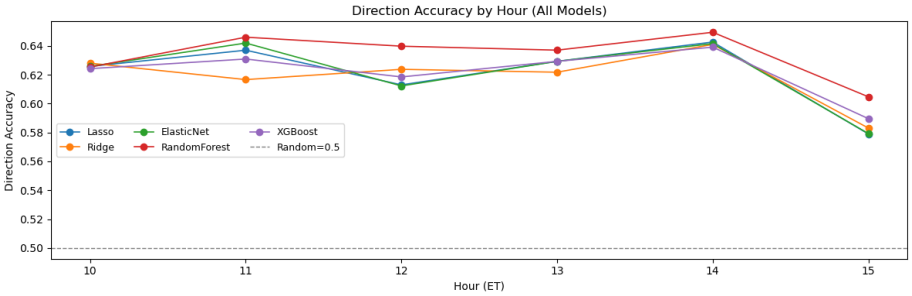


Figure 7: Accuracy

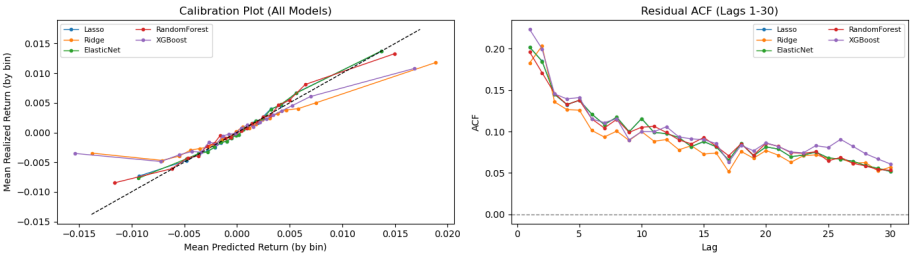


Figure 8: Enter Caption

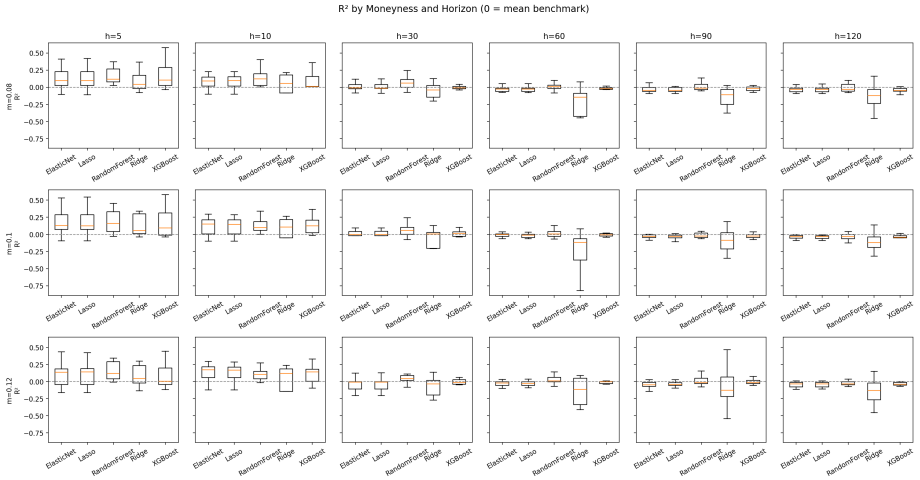


Figure 9: R2 BOX Plot

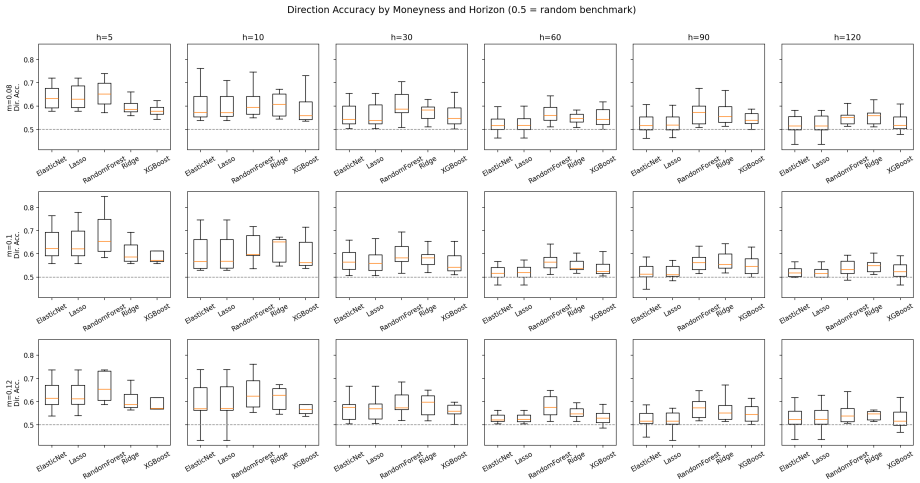


Figure 10: Driection ACC box plot

Table 3: Main summary across different minimum sample thresholds ($\min_n$).

$\min_n$	Model	Eq. Avg. R^2	Eq. Avg. Dir. Acc.	$R^2 > 0$ Share	Win Rate (R^2)	Win Rate (Dir. Acc.
20	Lasso	-0.0206	0.5532	0.3873	0.0857	0.0515
	Ridge	-0.1023	0.5737	0.3929	0.1841	0.2865
	ElasticNet	-0.0208	0.5531	0.3841	0.1214	0.0423
	RandomForest	**0.0083**	**0.5735**	**0.5079**	**0.4468**	**0.5398**
	XGBoost	-0.0413	0.5486	0.3857	0.1619	0.0799
50	Lasso	-0.0133	0.5497	0.4012	0.1032	0.0415
	Ridge	-0.1056	0.5673	0.4219	0.1245	0.2540
	ElasticNet	-0.0155	0.5498	0.3954	0.1289	0.0232
	RandomForest	**0.0261**	**0.5840**	**0.5474**	**0.4724**	**0.5934**
	XGBoost	-0.0146	0.5590	0.4057	0.1709	0.0880
100	Lasso	0.0112	0.5529	0.4182	0.0960	0.0436
	Ridge	-0.1089	0.5717	0.3944	0.0838	0.2374
	ElasticNet	0.0113	0.5525	0.4076	0.1303	0.0209
	RandomForest	**0.0566**	**0.5876**	**0.6308**	**0.5909**	**0.6196**
	XGBoost	0.0127	0.5638	0.4136	0.0990	0.0785

Notes: Results are equally weighted across underlyings.

Table 4: Performance by prediction horizon.

Horizon	Model	Eq. Avg. R^2	Eq. Avg. Dir. Acc.
5	Lasso	0.1496	0.6337
	Ridge	-0.0028	0.6037
	ElasticNet	0.1494	0.6316
	RandomForest	**0.1854**	**0.6721**
	XGBoost	0.1354	0.6027
10	Lasso	0.0546	0.6036
	Ridge	-0.2679	0.6113
	ElasticNet	0.0576	0.6037
	RandomForest	**0.1415**	**0.6268**
	XGBoost	-0.0612	0.5904
30	Lasso	-0.0138	0.5663
	Ridge	-0.0595	0.5811
	ElasticNet	-0.0132	0.5680
	RandomForest	**0.0573**	**0.5999**
	XGBoost	-0.0124	0.5631
60	Lasso	-0.0286	0.5343
	Ridge	-0.1610	0.5536
	ElasticNet	-0.0272	0.5335
	RandomForest	**0.0287**	**0.5787**
	XGBoost	-0.0459	0.5372
90	Lasso	-0.0119	0.5346
	Ridge	-0.0905	0.5698
	ElasticNet	-0.0116	0.5369
	RandomForest	**0.0275**	**0.5832**
	XGBoost	0.0052	0.5569
120	Lasso	-0.0522	0.5143
	Ridge	-0.1176	**0.5450**
	ElasticNet	-0.0545	0.5138
	RandomForest	**-0.0324**	0.5420
	XGBoost	-0.0450	0.5319

Notes: Results are equally weighted across underlyings.

References

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